Answer these Questions using CTE

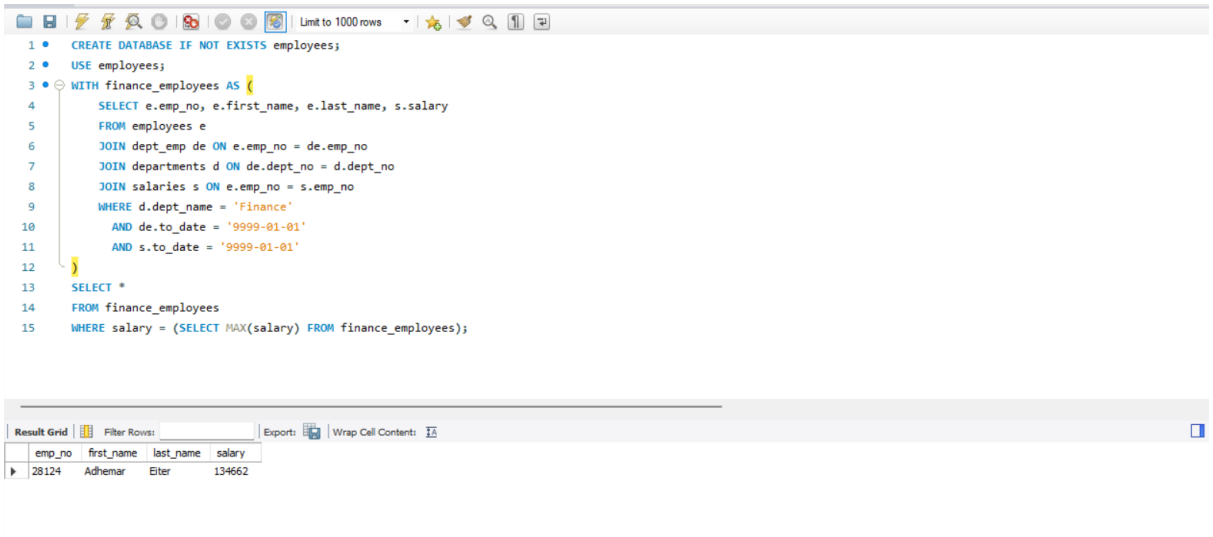
Used chatgpt for syntax correctness when required

1.Find the highest salaried employee of finance department.

Query:

```
WITH finance_employees AS (
   SELECT e.emp_no, e.first_name, e.last_name, s.salary
   FROM employees e
   JOIN dept_emp de ON e.emp_no = de.emp_no
   JOIN departments d ON de.dept_no = d.dept_no
   JOIN salaries s ON e.emp_no = s.emp_no
   WHERE d.dept_name = 'Finance'
     AND de.to_date = '9999-01-01'
     AND s.to_date = '9999-01-01'
)
SELECT *
FROM finance_employees
WHERE salary = (SELECT MAX(salary) FROM finance_employees);
```

Output:

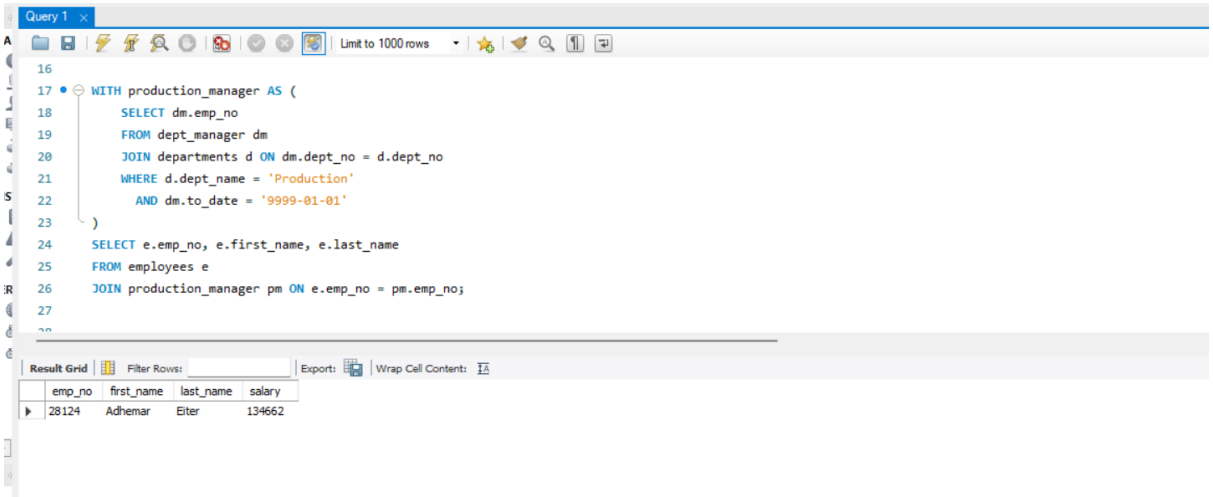


2.Current manager of the production department.

Query:

```
WITH production_manager AS (
   SELECT dm.emp_no
   FROM dept_manager dm
   JOIN departments d ON dm.dept_no = d.dept_no
   WHERE d.dept_name = 'Production'
     AND dm.to_date = '9999-01-01'
)
SELECT e.emp_no, e.first_name, e.last_name
FROM employees e
JOIN production_manager pm ON e.emp_no = pm.emp_no;
```

Output:

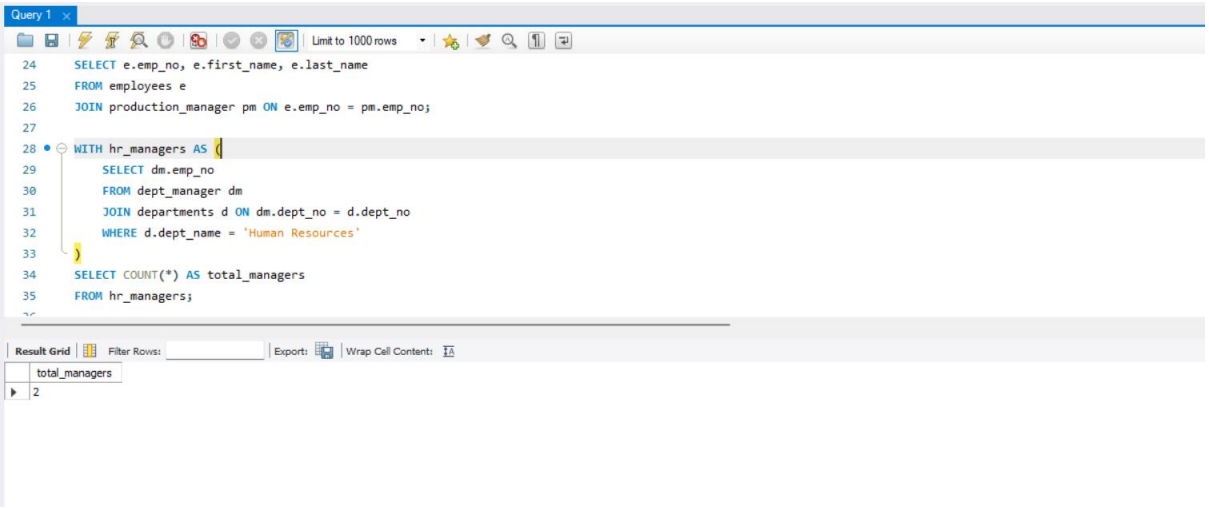


3.How many managers are worked in the Human Resources department.

Query:

```
WITH hr_managers AS (
   SELECT dm.emp_no
   FROM dept_manager dm
   JOIN departments d ON dm.dept_no = d.dept_no
   WHERE d.dept_name = 'Human Resources'
)
SELECT COUNT(*) AS total_managers
FROM hr_managers;
```

Output:



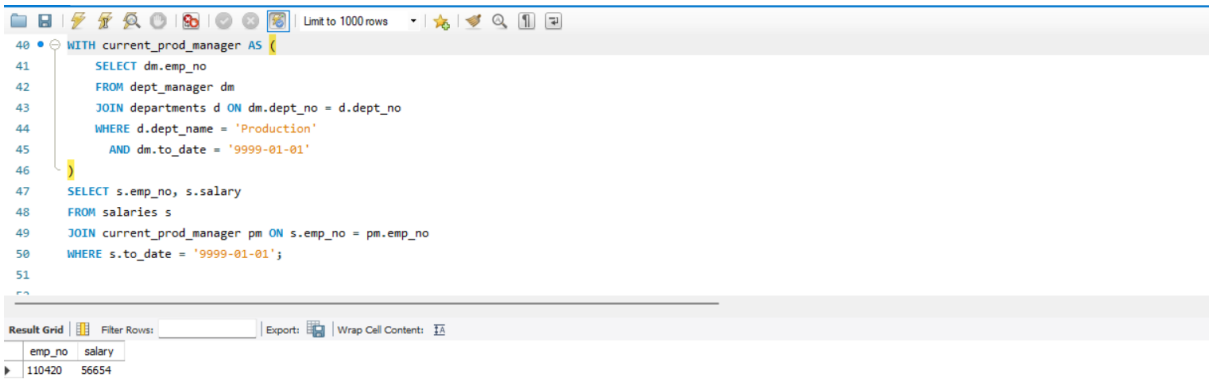
4.Salary of the Current Manager of the Production Department.

Query:

```
WITH current_prod_manager AS (
   SELECT dm.emp_no
   FROM dept_manager dm
   JOIN departments d ON dm.dept_no = d.dept_no
   WHERE d.dept_name = 'Production'
```

```
    AND dm.to_date = '9999-01-01'
)
SELECT s.emp_no, s.salary
FROM salaries s
JOIN current_prod_manager pm ON s.emp_no = pm.emp_no
WHERE s.to_date = '9999-01-01';
```

Output:

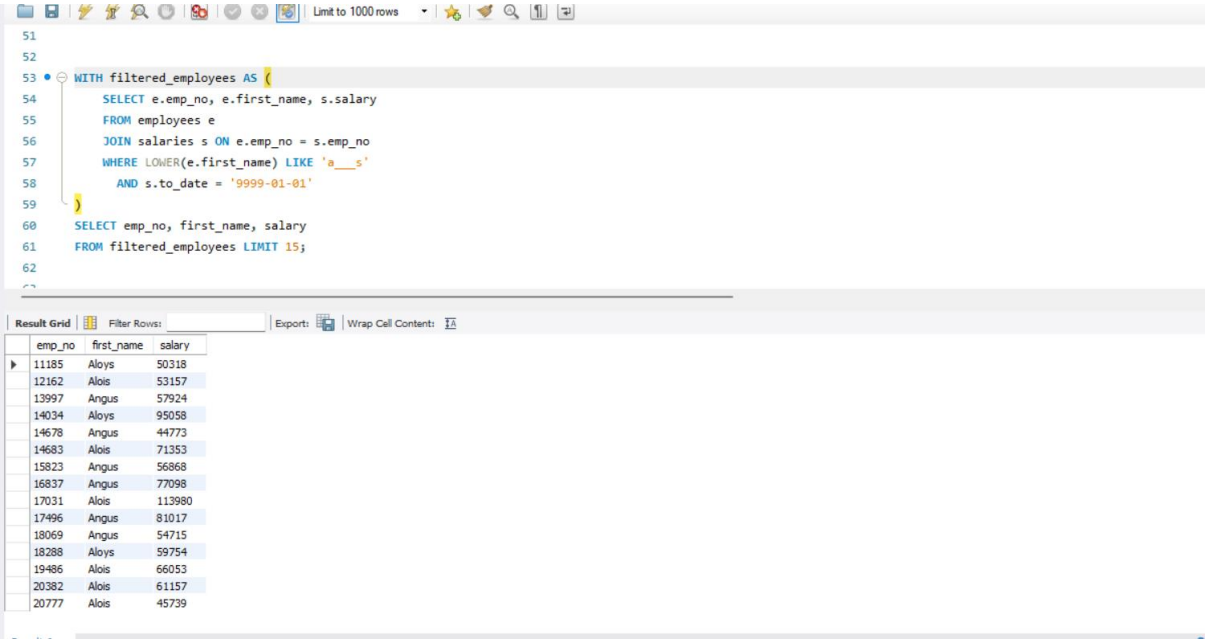


5.Find the people whose name starts with a and end with s with only 5 letters , and their salaries.

Query:

```
WITH filtered_employees AS (
   SELECT e.emp_no, e.first_name, s.salary
   FROM employees e
   JOIN salaries s ON e.emp_no = s.emp_no
   WHERE LOWER(e.first_name) LIKE 'a___s'
     AND s.to_date = '9999-01-01'
)
SELECT emp_no, first_name, salary
FROM filtered_employees LIMIT 15;
```

Output:



```
WITH filtered_employees AS (
    SELECT e.emp_no, e.first_name, s.salary
    FROM employees e
    JOIN salaries s ON e.emp_no = s.emp_no
    WHERE LOWER(e.first_name) LIKE 'a___s'
      AND s.to_date = '9999-01-01'
)
SELECT emp_no, first_name, salary
FROM filtered_employees LIMIT 15;
```

emp_no	first_name	salary
11185	Aloys	50318
12162	Alois	53157
13997	Angus	57924
14034	Aloys	95058
14678	Angus	44773
14683	Alois	71353
15823	Angus	56868
16837	Angus	77098
17031	Alois	113980
17496	Angus	81017
18069	Angus	54715
18288	Aloys	59754
19486	Alois	66053
20382	Alois	61157
20777	Alois	45739

6.Find the employees whose hire date and From date are different ,how much difference is there for each one.

Query:

```
WITH date_diff AS (
   SELECT e.emp_no,
          e.hire_date,
          de.from_date,
          DATEDIFF(de.from_date, e.hire_date) AS date_difference
   FROM employees e
   JOIN dept_emp de ON e.emp_no = de.emp_no
)
SELECT *
FROM date_diff
WHERE hire_date <> from_date LIMIT 15;
```

Output:

```
WITH date_diff AS (
    SELECT e.emp_no,
           e.hire_date,
           de.from_date,
           DATEDIFF(de.from_date, e.hire_date) AS date_difference
    FROM employees e
    JOIN dept_emp de ON e.emp_no = de.emp_no
)
SELECT *
FROM date_diff
WHERE hire_date <> from_date LIMIT 15;
```

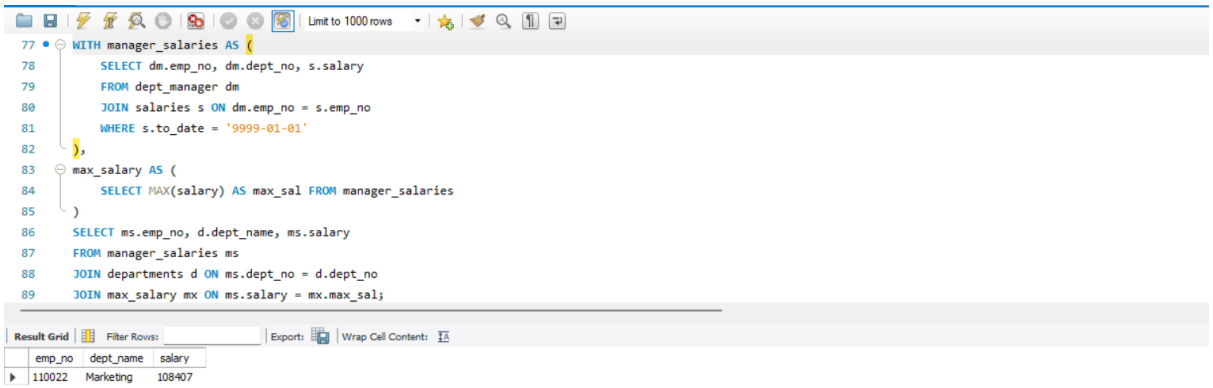
emp_no	hire_date	from_date	date_difference
10002	1985-11-21	1996-08-03	3908
10003	1986-08-28	1995-12-03	3384
10006	1989-06-02	1990-08-05	429
10008	1994-09-15	1998-03-11	1273
10010	1989-08-24	1996-11-24	2649
10010	1989-08-24	2000-06-26	3959
10014	1987-03-11	1993-12-29	2485
10015	1987-07-02	1992-09-19	1906
10016	1995-01-27	1998-02-11	1111
10018	1987-04-03	1992-07-29	1944
10020	1991-01-26	1997-12-30	2530
10022	1995-08-22	1999-09-03	1473
10023	1989-12-17	1999-09-27	3571
10024	1997-05-19	1998-06-14	391
10027	1989-07-07	1995-04-02	2095

7.Highest salaried manager and for which department he is present in, i need dept name ,and manager emp_no as output

Query:

```
WITH manager_salaries AS (
   SELECT dm.emp_no, dm.dept_no, s.salary
   FROM dept_manager dm
   JOIN salaries s ON dm.emp_no = s.emp_no
   WHERE s.to_date = '9999-01-01'
),
max_salary AS (
   SELECT MAX(salary) AS max_sal FROM manager_salaries
)
SELECT ms.emp_no, d.dept_name, ms.salary
FROM manager_salaries ms
JOIN departments d ON ms.dept_no = d.dept_no
JOIN max_salary mx ON ms.salary = mx.max_sal;
```

Output:



8.Find which current department manager has less salary than highest salaried employee of that department.

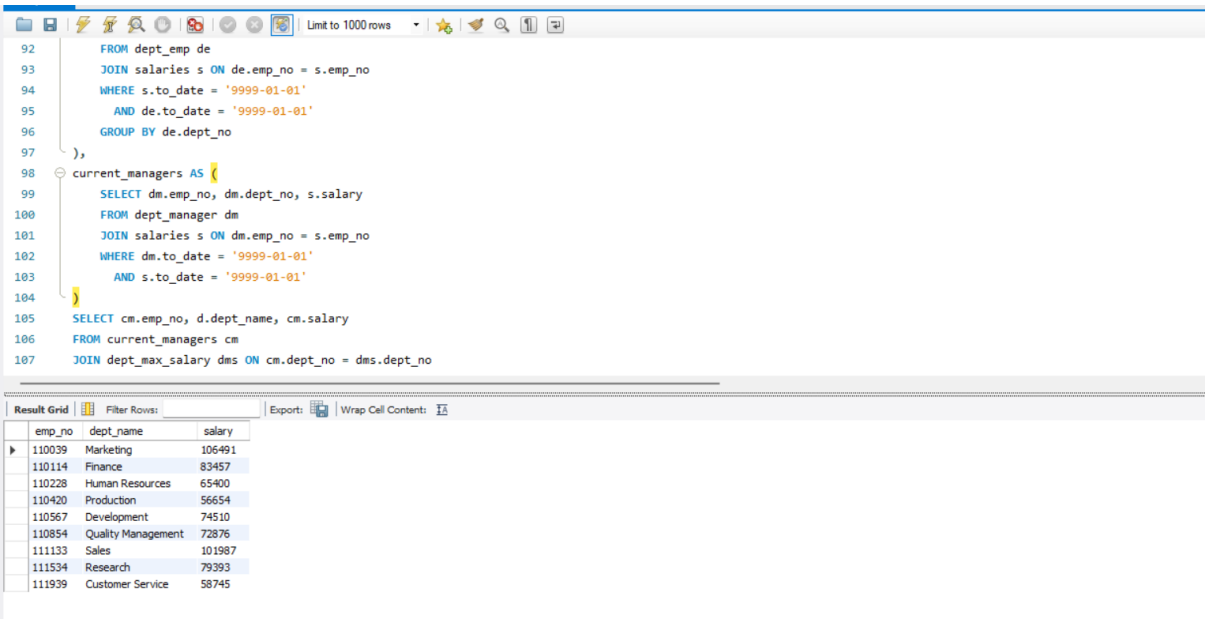
Query:

```
WITH dept_max_salary AS (
   SELECT de.dept_no, MAX(s.salary) AS max_emp_salary
   FROM dept_emp de
   JOIN salaries s ON de.emp_no = s.emp_no
   WHERE s.to_date = '9999-01-01'
     AND de.to_date = '9999-01-01'
   GROUP BY de.dept_no
),
current_managers AS (
   SELECT dm.emp_no, dm.dept_no, s.salary
   FROM dept_manager dm
   JOIN salaries s ON dm.emp_no = s.emp_no
   WHERE dm.to_date = '9999-01-01'
     AND s.to_date = '9999-01-01'
)
SELECT cm.emp_no, d.dept_name, cm.salary
FROM current_managers cm
JOIN dept_max_salary dms ON cm.dept_no = dms.dept_no
```

JOIN departments d ON cm.dept_no = d.dept_no

WHERE cm.salary < dms.max_emp_salary;

Output:



```
92      FROM dept_emp de
93      JOIN salaries s ON de.emp_no = s.emp_no
94      WHERE s.to_date = '9999-01-01'
95        AND de.to_date = '9999-01-01'
96      GROUP BY de.dept_no
97  ),
98  current_managers AS (
99      SELECT dm.emp_no, dm.dept_no, s.salary
100     FROM dept_manager dm
101     JOIN salaries s ON dm.emp_no = s.emp_no
102     WHERE dm.to_date = '9999-01-01'
103       AND s.to_date = '9999-01-01'
104 )
105 SELECT cm.emp_no, d.dept_name, cm.salary
106 FROM current_managers cm
107 JOIN dept_max_salary dms ON cm.dept_no = dms.dept_no
```

emp_no	dept_name	salary
110039	Marketing	106491
110114	Finance	83457
110228	Human Resources	65400
110420	Production	56654
110567	Development	74510
110854	Quality Management	72876
111133	Sales	101987
111534	Research	79393
111939	Customer Service	58745

9.Which department manager has the highest salary among all departments. output : emp_no ,salary,dept_name

Query:

```
WITH manager_salary AS (
   SELECT dm.emp_no, dm.dept_no, s.salary
   FROM dept_manager dm
   JOIN salaries s ON dm.emp_no = s.emp_no
   WHERE dm.to_date = '9999-01-01'
     AND s.to_date = '9999-01-01'
),
highest_manager AS (
   SELECT MAX(salary) AS max_salary FROM manager_salary
)
SELECT ms.emp_no, ms.salary, d.dept_name
```

FROM manager_salary ms

JOIN highest_manager hm ON ms.salary = hm.max_salary

JOIN departments d ON ms.dept_no = d.dept_no;

Output:

```
WITH manager_salary AS (
    SELECT dm.emp_no, dm.dept_no, s.salary
    FROM dept_manager dm
    JOIN salaries s ON dm.emp_no = s.emp_no
    WHERE dm.to_date = '9999-01-01'
      AND s.to_date = '9999-01-01'
),
highest_manager AS (
    SELECT MAX(salary) AS max_salary FROM manager_salary
)
SELECT ms.emp_no, ms.salary, d.dept_name
FROM manager_salary ms
JOIN highest_manager hm ON ms.salary = hm.max_salary
JOIN departments d ON ms.dept_no = d.dept_no;
```

Result Grid

emp_no	salary	dept_name
110039	106491	Marketing